Rendering Soundscapes

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Abstract

The recording of an aural soundscape can be used as a structural framework in which the composer can reconceptualize and reshape the entire space. Using an eco-structuralist process the recording is reduced to a framework that affords an entirely new sonic perception, whilst retaining a sense of the original space. This paper discusses the framework and rendering of a seaside sound event.

Introduction

Eco-structuralism contains a set of rules and techniques that can be employed when working with recorded sounds. The techniques are designed to extract and maintain the structure of a recorded sound so that it may be used in a number of different compositional forms. In this paper the techniques will be used on a common seaside sound event to reshape the soundscape. The process and results will be analysed with a focus on the continuity and discontinuity of the sonic space.

Extracting the Structure

Eco-structuralism is a compositional technique that employs very strict rules for extraction and processing of sound event structures (Opie & Brown 2006). The rules are designed with persistence of the structures as a primary outcome. The persistent nature of the structure affords an understanding of the original encapsulating sound event, whilst also allowing the new schizophrenic[1] event to generate its own generational identity (Gibson 1966, Schafer 1994). Eco-structuralism attempts to interact with the sound event in a non-trivial way (Dunn 1984).

> By zeroing in on the structure of the sound event itself, it becomes possible to extend the network of meanings along most time scales and into the realm of perceptual cues. (Keller 2006)

The structure used in this paper is a seaside sound event containing a common ocean wave and swell keynote sound of many beaches without any discernible soundmark. Because the process requires sonic audition to make an informed decision on the process, this paper is attempting to represent some of the audible ideas graphically by applying similar techniques to a related image. Represented first in Figures 1 & 2 are the original source files. The image is just a quick snapshot of the space, representing only a few milliseconds of the event, whereas the recording represents 25 seconds of the event as it took place.



Figure 1. *Image of the Ninety Mile Beach.*

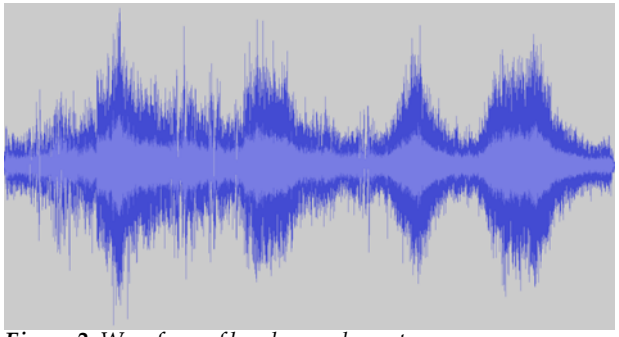


Figure 2. *Waveform of beach sound event*

The simplest structure to extract is the amplitude envelope. *ExtractAmplitude.java* performs an amplitude attribute study that analyses and extracts the peak amplitudes (Opie 2005). This program generates an XML document containing information on the audio file, including a description, file formatting, analysis length, samples taken, minimum and maximum values, links to the audio file and extra comments, and most importantly it lists the peak amplitudes. Being in an XML format the information is easily interoperable with other programs, and easily accessible. Figure 3 shows the beach image with the outlines highlighted, and the rest of the image information falling into obscurity. This is essentially what the composer is left with after performing an amplitude extraction process. All that is left is an envelope, which contains information on at-

[1] It is acknowledged that the sound is already schizophrenic, having been removed from its originating source and place in time, however this new process creates a much deeper level of schizophrenia.

tack peaks, sustains, decays and releases. Comparing Figure 2 to Figure 4, it is clear to see the correlation between the waveform of Figure 2, and the peak amplitudes of the graph in Figure 4. Audio tests with these amplitude structures have shown just how important the amplitude envelope is in determining the origin of the sound event (Opie 2005).



Figure 3. *Image of the Ninety Mile Beach showing just outlines – comparable to amplitude structure.*

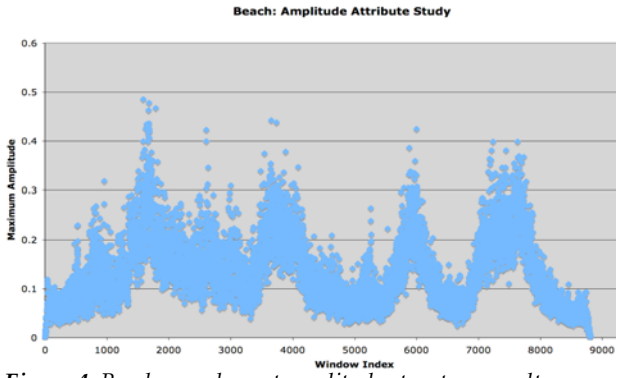


Figure 4. *Beach sound event amplitude structure results graphed.*

The second structure to be extracted is the timbre. Following the timbre attribute study *ExtractTimbre.java* performs an FFT analysis and determines ratios that define the timbral structure of the beach event. This information is collected in an XML document that is structured in the same way as the amplitude XML document. The timbre structure contains a coarse structure of relations between layered sounds within the event. The best graphical representation of the structure would be that of Figure 5 in which the outlines are becoming blurred, and textures are looking more like coarse grains. The timbre attribute study seeks to set up ratios by which a similar timbred sound can be created by applying the ratios to its fundamental frequency.

The timbre attribute study still needs some refinement, which may reduce the coarseness of the structure, although this may come at the cost of a ridiculously large data file. There are some avenues of investigation that need to be followed in order to improve this process, but these will not be discussed here. Figure 6 shows a spectrograph of the beach event, from which the timbral data is gathered. As can be seen from this figure, the timbre becomes very rich and noisy as the waves crash on the shore. There is a high correlation between the amplitude and timbral richness that can be utilised and exploited with these structures. For example a sound model of a beach sound event could be emulated effectively by increasing the timbral richness proportionally to the amplitude increase. Such a model would operate entirely by manipulation of the amplitude, similar to the crackling fire model demonstrated by Opie (2005).



Figure 5. *Image of the Ninety Mile Beach with the texture amplified – comparable to timbre structure.*

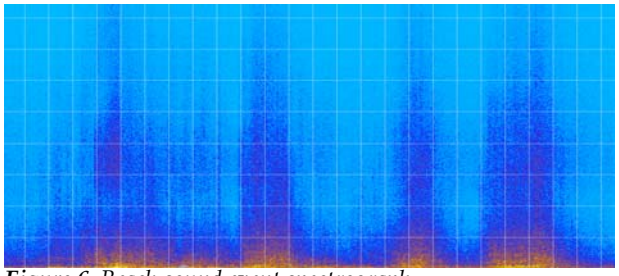


Figure 6. *Beach sound event spectrograph.*

The third structure to be extracted from the seaside event is the fundamental frequency. *ExtractFrequency.java* calculates and extracts the fundamental frequency, using a fundamental frequency attribute study (Opie 2006). The results are again recorded in an XML document that is structured in the same way as the amplitude document. The fundamental frequency study seeks to find the base frequency upon which all other harmonics rest. This is achieved by finding the significant-zero-crossing (SZC) and calculating the frequency based on the period. The SZC is determined by the peak amplitude value, of the waveform. The peak must fall within the mean range threshold of previous peaks. This process reduces multiple crossings that occur due to minor fluctuations that occur around the crossing point. Once the fundamental frequency is known, it creates a place with which to employ the timbre structure. Applying the amplitude structure to the combined frequency and timbre structure gives the resulting sound a sharper contour.

The fundamental frequency extraction process itself would be best visualised by slightly pixelating the image, as in Figure 7. There is a significant loss of timbral richness from the sound event, the base tone being the only feature retained. The picture can still be rebuilt with that knowledge, but it would look more like a paint-by-number exercise than a complete artwork. Figure 8 displays a graph of the fundamental

frequency. Most frequencies stay under 3000Hz, although there are a few anomalies reaching as high as 8000Hz during the crashing of the wave, which indicates some fluctuation of the signal. The lowest portion of the graph is most telling, as there is a definite rise in frequency that coincides with each crash of the waves.

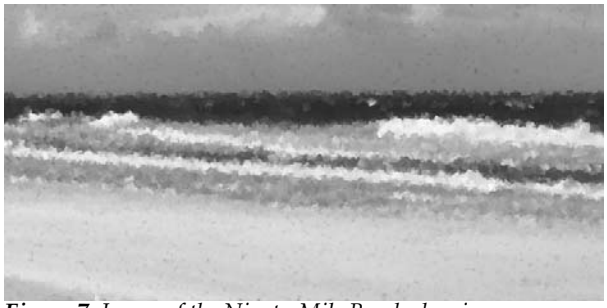


Figure 7. *Image of the Ninety Mile Beach showing average colour segregated into small windows – comparable to fundamental frequency structure.*

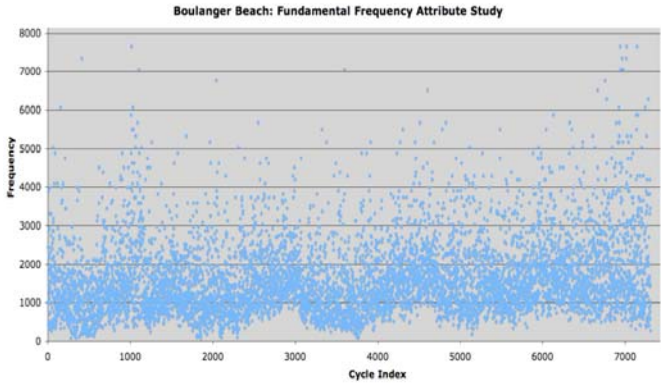


Figure 8. *Beach sound event fundamental frequency structure results graphed.*

A spatial attribute study could also be made, but as this is a mono recording such a test would reveal nothing. A durational attribute study has also been considered as the next step, but it has not been developed yet, although an approximation of the results reveal that the information is also captured in the timbre structure, although not yet in a singular accessible format.

Processing the Structure

With these three structures a new soundscape can now be rendered. The structures are like any building structure. They constitute the framework. Eco-structuralism allows for the rotation, stretching and compression of these structures (Opie & Brown 2006), although for this exercise they will be left unchanged, acting entirely as the framework from which they originated. The first rendering will impose a bell ringing upon the beach sound event. The bell ring contains a full rich tone, which can easily fill out the structures, in terms of amplitude and spectral detail. The fundamental frequency structure will cause a varying transposition throughout the bell ring, causing a warping of the sound. The timbre structure will also transpose the timbre of the bell ring to suit that of the beach, and the amplitude structure will give the bell a new dynamic shape. Figure 9 depicts this process as adding a chrome finish to the beach image structure. There is definitely some transposition and rippling of the image, as well as new content added to the image. The beach form can still be made out, but it has an entirely new rendered façade. The sonic space is the same, although the subject has been imbued with a new entity. Because the beach structure is still intact, the perception of the beach should also remain somewhat intact, or it should at least afford the notion of a beach, as well as that of a bell.

Visually this has been captured. Figure 10 is the waveform of the new soundscape. As can be seen it is very similar to the original waveform. On listening to the new soundscape the perception of the beach is still present, the bell just creating a new perspective on the beach.



Figure 9. *Image of the Ninety Mile Beach structure with a chrome finish – comparable to a using a rich tone as the content within which to fill the structure.*

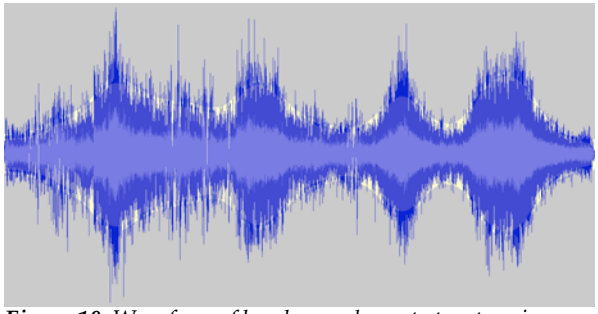


Figure 10. *Waveform of beach sound event structure infused with a rich tone.*

The Bell demonstrates this simple experiment effectively, but what if a much more contrasting sound is used as the new structure filler? What happens when you mix water and fire? As with the previous example the structure itself will not be transformed, it will act only as a direct framework in order to create a balanced juxtaposition. Visually the beach structure in Figure 11 has been filled with a dried earth image, and a stone wall for the sky, as this also seemed an appropriate visual contrast. The dry land has water washing over it, with a brick wall the horizon. The perception of the beach still exists, but the contrast between wet and dry causes the viewer to re-evaluate the entire image.

For an audio example a fire sound event works much better than the recording of dry

land. A campfire sample has been placed within the confines of the beach structure. The fire does not fill the space as readily as the bell. It leaves gaps in the rendered walls of the structure. The spectrum of the fire easily fills the beach structure, but the amplitude structure is much larger than that of the fire.

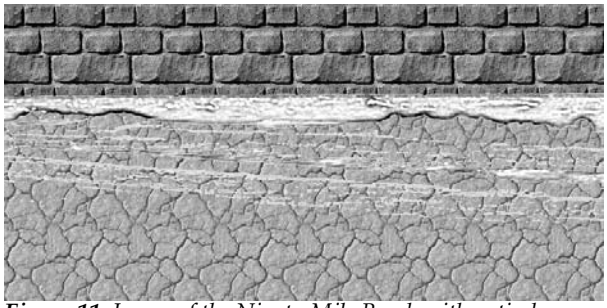


Figure 11. *Image of the Ninety Mile Beach with entirely new images built within the structure - comparable to juxtaposing sound events created within the confines of the sound structure.*

The recording of the fire is clearly discernable within the confines of the beach structure, but the beach structure can still be perceived. Water and fire can coexist peacefully, although not without causing the listener to re-evaluate what they think they are hearing. The affordances of water and fire would not have many overlapping features, they would instead cancel each other out more often than not.

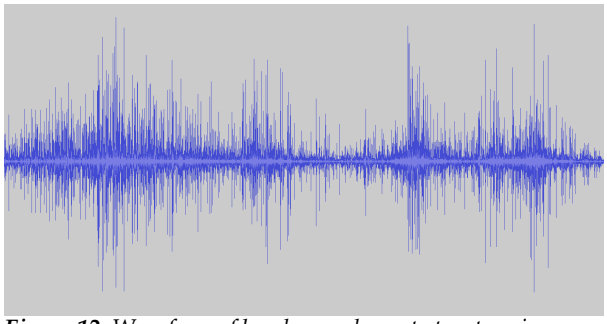


Figure 12. *Waveform of beach sound event structure infused with a crackling fire sound event.*

Future Structural Devices

A couple of future structural directions have already been mentioned in this paper, namely creating a duration attribute study, and also a beach-modelling project in which the beach sound event is physically modelled and emulated. Also the notion of structural transformation has also been raised. For this paper the space remained the same, whilst the contents changed. It would be interesting to examine how persistent the beach was after a number of structural transformations, especially if those transformations were quite radical. Eco-structuralism allows for quite a number of transformations to take place so long as the data remains in serial unity.

Other avenues that need investigating are the psychological ramifications of the extraction of structures, transformation and juxtaposition processes. Whilst the structures can be perceived, are they still strong enough to afford the same response as would be received from the original recording (Dunn 2001)? Also if the sounds are used in the described way, would they promote better listeners? If the audience needs to re-evaluate all the sounds they are hearing would they listen more closely? If they are listening more closely to the newly rendered soundscapes, does that mean they would listen more closely to the real sounds in their environment (Schafer 1994)?

Conclusion

This paper has explored the extraction of a number of structures from a beach sound event, discussing some of the potential outcomes of these structures. It has then used the structures as a framework within which other sounds events were placed. The paper has shown that the structures can still be perceived despite a radical change of audio content. The perception of the two juxtaposing sound events causes re-evaluation and perhaps more meaningful listening of the newly rendered soundscape.

References

Dunn, D. 2001. *Santa Fe Institute Public Lecture*, http://www.davidddunn.com/~david/writings/sfilecture.PDF Accessed 16/09/2007.

Dunn, D. 1984. *Music, Language, And Environment*. http://www.davidddunn.com/~david/writings/mle.pdf Accessed: 16/09/2007.

Gibson, J.J. 1966. *The Senses Considered as Perceptual Systems*. Boston, MA: Houghton Mifflin.

Keller, D. and Capasso, A. 2006. "New Concepts and Techniques in Eco-Composition", *Organized Sound*. 11(1). Cambridge, UK. University Press.

Keller, D. 2004. *Paititi: A multimodal journey to El Dorado*. Doctor in Musical Arts Thesis. Deparment of Music, Stanford University. CA, USA.

Kivy, P. 2002. *Introduction to a Philosophy of Music*. Clarendon Press.

Opie, T. and Brown, A. R. 2006. "An Introduction to Eco-Structuralism", *Proceedings of the International Computer Music Conference*. New Orleans, USA. ICMA, pp. 9-12.

Opie. T. 2006. "Fundamental Frequency as an Attribute of Eco-Structualism", *Proceedings of the Australasian Computer Music Conference*, Adelaide, Australia.

Opie. T. 2005. "Amplitude Analysis as Musical Composition", *Proceedings of the Australasian Computer Music Conference*, Brisbane, Australia.

Schafer, R. M. 1994. *The Soundscape: Our Sonic Environment and the Tuning of the World*. Destiny Books.

Xenakis, I. 1971. *Formalized Music*. Indiana University Press, Bloomington, USA.